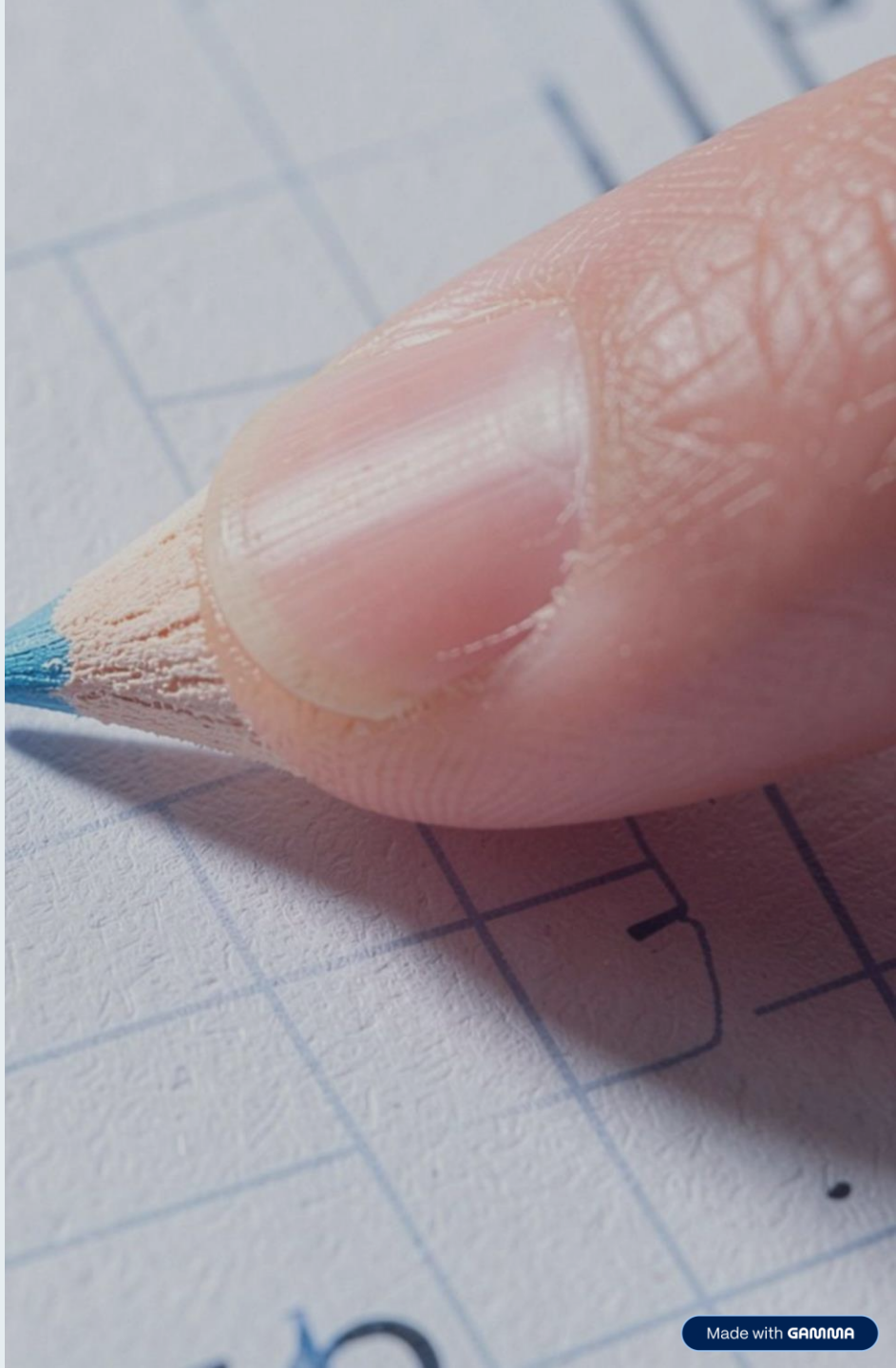
DSA0402 FUNDAMENTALS OF DATA SCIENCE

Linear Regression

Gayathri. M (192472074)
Jannathul Firdaus. N (192573025)
Nandhana Sri. S (1972260063)



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THE CORE IDEA

Linear regression connects data to predictions

A simple relationship can become a practical estimate.



Finds a relationship

It connects an input with an output.



Summarizes the pattern

A straight line captures the overall direction.



Predicts the unknown

Known data helps estimate a new value.

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INPUT → MODEL → OUTPUT

Two variables define the prediction

One variable provides information. The other is the value we want to explain or predict.



Example: study hours (X) can help explain marks (Y).

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MODEL TYPES

Two forms cover most use cases

Both estimate a numeric output.

Simple

One input predicts one output.

`Study hours → marks`

Multiple

Several inputs predict one output.

`Hours + attendance + assignments → marks`

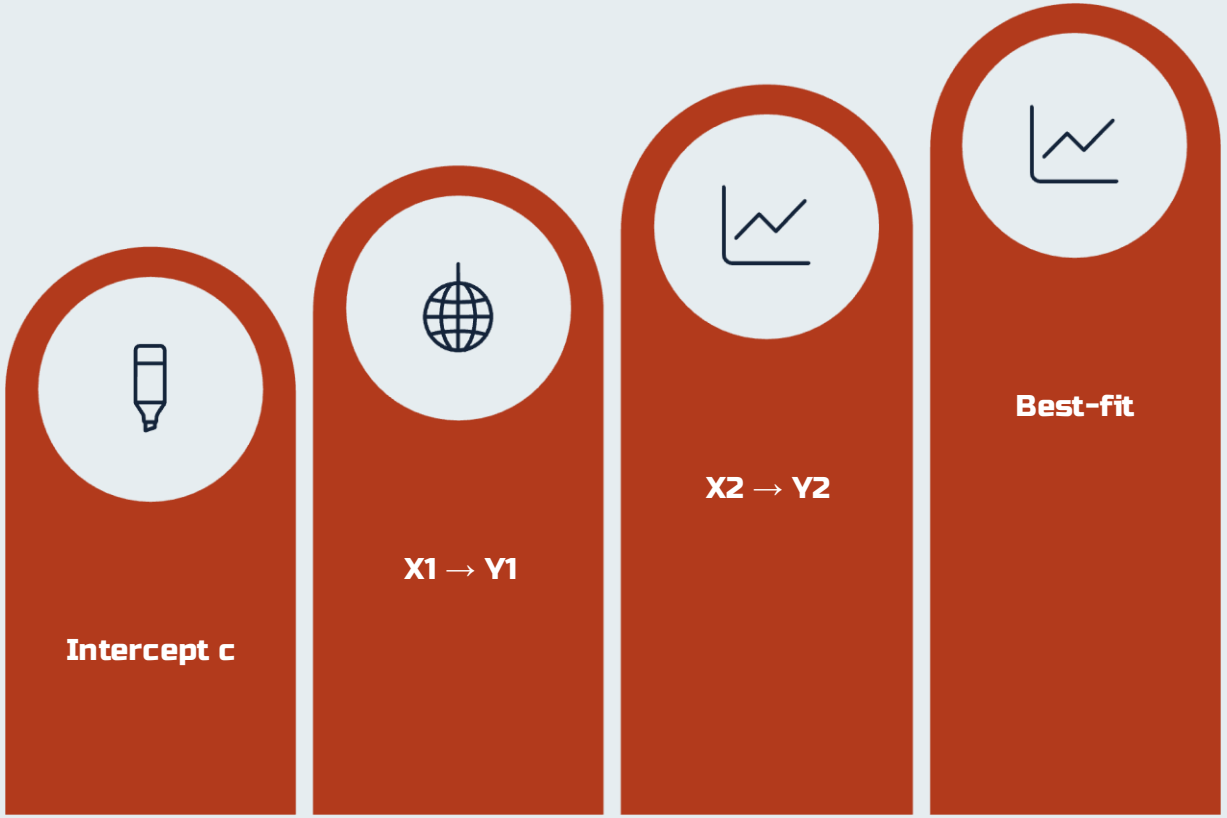
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THE MODEL IN ONE LINE

Y = mX + c

Y is the predicted output. **X** is the input. **m** is the slope, and **c** is the intercept.

The slope changes Y for each unit of X. The intercept is predicted Y when X = 0.



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FROM DATA TO MODEL

Training follows four clear steps

Collect

A dataset with examples.

Identify

Separate input X from output Y.

Fit

Reduce prediction error.

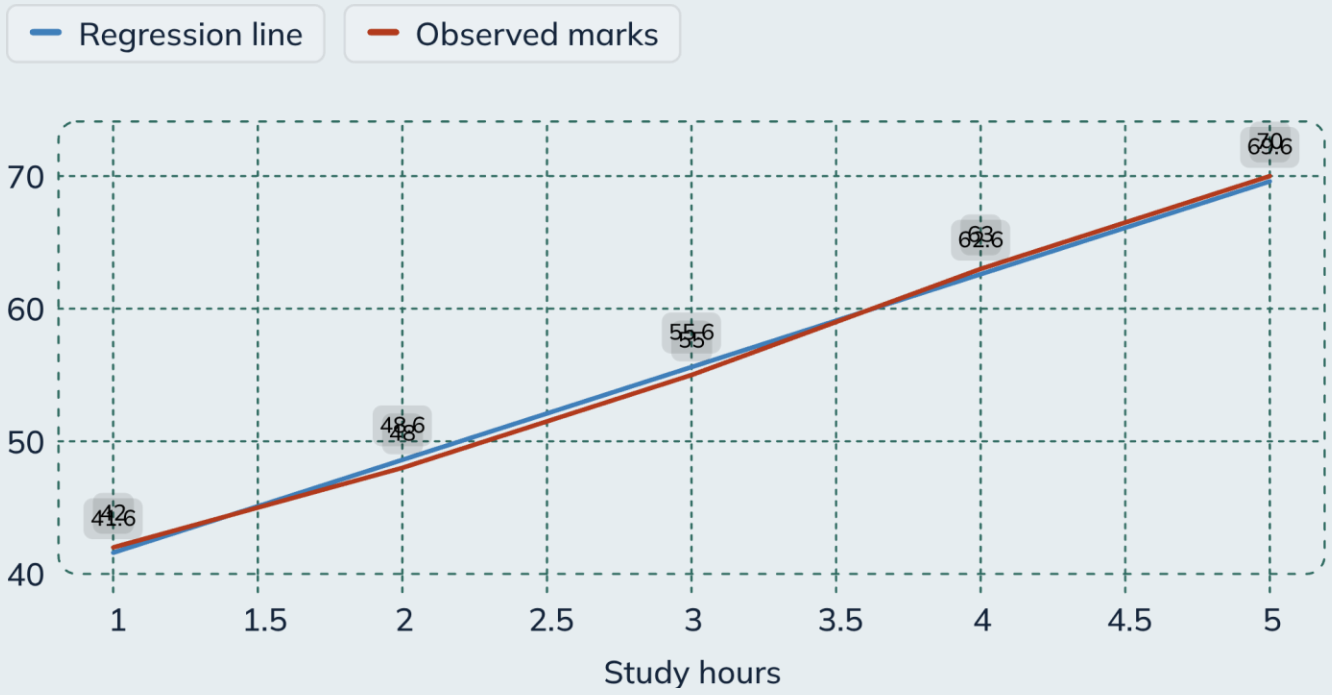
Predict

Apply the trained model to new values.

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WORKED EXAMPLE

Study hours can estimate student marks



Sample data

Hours = [1, 2, 3, 4, 5]

Marks = [42, 48, 55, 63, 70]

For a new number of study hours, the fitted line gives an estimated mark.

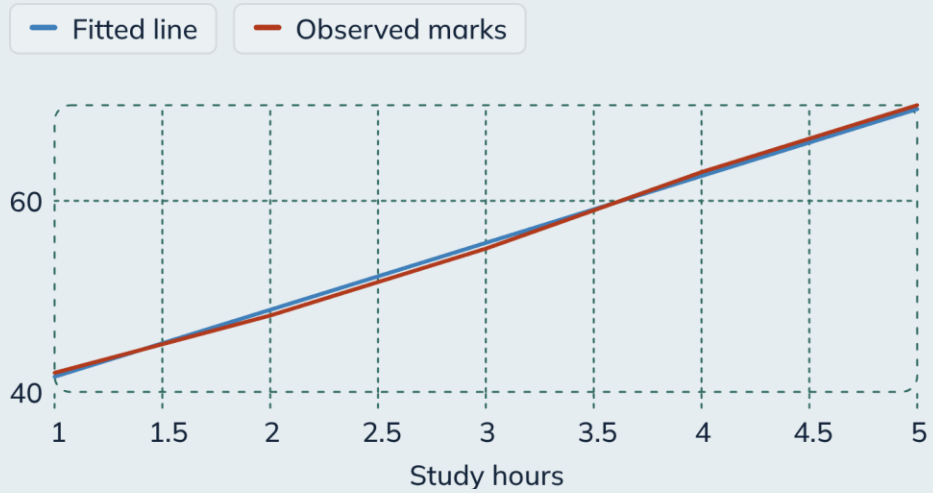
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A SMALL PYTHON WORKFLOW

Python makes the model practical

Use **pandas** for data, **matplotlib** for plots, and **scikit-learn** for the model.

```
from sklearn.linear_model import
LinearRegression
X = [[1], [2], [3], [4], [5]]
y = [42, 48, 55, 63, 70]
model = LinearRegression().fit(X, y)
print(model.predict([[6]]))
```



The same data, now fitted and ready for a new prediction.

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WHEN TO USE IT

Linear regression is useful, but not universal

Advantages

- Simple to understand and implement
- Fast for small and large datasets
- Coefficients are easy to interpret

Limitations

- Sensitive to outliers
- Assumes a roughly linear relationship
- Performs poorly on strongly nonlinear patterns

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FINAL TAKEAWAY

Start with the relationship, then test the prediction.



Inspect

Summarize how variables move together.



Model

Support prediction and data analysis with an interpretable model.



Test

Inspect data, assumptions, and errors before trusting the result.

A simple line can be a powerful first model. Thank you.

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